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P

COMMON QUARTERLY EXAMINATION - 2025

Standard - 12 Reg.No. | | | | | | |

BIOLOGY

Time Allowed:3.00 Hours *Maximum Marks:70*

SECTION - A (MARKS : 35)

BIO-BOTANY

PART-I **8×1=8**

Note : i) Answer all the questions:
ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.

1) The first cell of male gametophyte in angiosperm is _____.
a) Nucleus b) Microspore
c) Primary Endosperm Nucleus d) Megaspore

2) The scar left by funiculus in the seed is
a) Epicotyl b) Taxmen c) Hilum d) Radicle

3) The phenotypic ratio of complementary genes
a) 9:7 b) 9:3:3:1 c) 9:3:4 d) 9:6:1

4) "Gametes are never hybrid". This is a statement of
a) Law of dominance b) Law of independent assortment
c) Law of segregation d) Law of random fertilization

5) The bacteria responsible for inducing tumours in several dicot plants are _____.
a) Candida Utilise b) Spirulina
c) Chlorella d) Agrobacterium tumifaciens

6) PBR.322.BR stands for
a) Plasmid Bacterial Recombination b) Plasmid Bacterial Replication
c) Plasmid Baltimore and Rodriguez d) Plasmid Bolivar and Rodriguez

7) The appropriate temperature used for cryopreservation
a) 196°C b) –196°C c) 100°C d) –100°C

8) An allohexaploidy contains
a) Six different genomes b) Six copies of three different genomes
c) Two copies of three different genomes d) Six copies of one genome

PART-II **4×2=8**

Note : Answer any four of the following questions:

9) What is cantharophily?
10) Give the names of the scientists who rediscovered Mendelism.
11) Name the chemicals used in gene transfer.
12) Write short notes on sonara - 64.
13) What is gene Mapping?
14) Define Explant.

PART-III **3×3=9**

Note : Answer any three of the following questions:
Question no.19 is compulsory.

15) Draw the structure of ovule and label it.
16) Define Genetics.
17) List any three significance of ploidy.
18) Compare the various types of Blotting techniques.
19) Define sterilization. How will you sterilize a culture room?

PART-IV **2×5=10**

Note : Answer all the questions:

20) a) Enumerate the characteristic features of Entomophilous flowers. **(OR)**
b) Explain incomplete dominance with an example.

21) a) What is single cell protein? Write its applications. **(OR)**
b) Explain the basic concept involved in plant tissue culture.

Kindly Send Me Questions & Answer Keys to Us: padasalai.net@gmail.com

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SECTION-B (Marks:35)
BIO-ZOOLOGY
PART-I

Note : i) Answer all the questions.
ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.

Choose the best answers given as below: 8×1=8

1) The mode of reproduction in bacteria is by
   a) Formation of gamers b) Endospore formation
   c) Conjugarion d) Zoospore formation
2) Find the wrongly matched pair. Cells / organs - Secretory Produce.
   a) Sertoli cells - Inhibin b) Leg dig cells - Testosterone
   c) Endometrium - Vesiculase d) Zoospore formation
3) Which one of the following groups includes sexally transmitted diseases caused by bacteria only?
   a) Syphilis, gonorrhoea and candidiasis
   b) Syphilis, gonorrhoea and trichomoriasis
   c) Syphilis, trichomoriasis and pediculosis
4) ABO blood group in men is controlled by
   a) Multiple allies b) Lethal genes c) Sex linked genes d) Y-linked genes
5) Which of the following is not required for any of the techniques of DNA finger printing available at present?
   a) Zinc finger analysis b) Restriction enzymes
   c) DNA-DNA hybridization d) Polymerase chain reaction
6) The golden age of reptiles was
   a) Mesozoic era b) Cenozoic era c) Paleozoic era d) Proterozoic era
7) Choose the correctly match pair.
   a) Amphetamines - Stimulent b) LSD - Narcotic
   c) Heroin - Psychotropic d) Benzodiazepine - Pain killer
8) Match the following and choosing correct answer:
   a) Athlete's foot - I. Protozoan microbe
   b) Malaria - II. Bacteria
   c) Liver hepatitis - III. fungi
   d) Premonia - IV. hepatitis B virus

PART-II

Note : Answer any four of the following questions: 4×2=8

9) Which type of reproduction is effective asexual or sexual and why?
10) List out embryonic membranes.
11) How are SID's transmitted?
12) What are holandric genes?
13) Why the human genome project is called a mege project?
14) What's meant by Atavistic organs?

PART-III

Note : Answer any three of the following questions: 3×3=9
Question no.19 is compulsory.

15) Draw a labelled sketch of a spermatozoan.
16) What are the process involved in Amniocentesis?
17) Give any 3 applications of DNA finger printing.
18) Explain the three major categories.In which for fossilization occur.
19) When does antibiotic resistance develop?

PART-IV

Note : Answer all the questions: 2×5=10

20) Write essay an various parthenogenesis. **(OR)**
    What are the applications of karyotyping?
21) Explain the structure of immune - globulin with suitable diagram. **(OR)**
    Write short notes on the following:
    a) Prrewer's yeast b) Idenovella sakaiensis c) Microbial fuel cells

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